

**Geoscientific Project: Mapping and Morphological Analysis of Bottom-
Current Controlled Sediment Waves in Contourite Drifts.**

Prepared

by:

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Abstract

Sediment waves are important geomorphic features that provide valuable insights into the dynamics of deep-water bottom currents and sediment transport processes. This study investigates the morphology of sediment waves in the Argentine Basin and the Hatteras Outer Ridge morphometric measurements. Morphological parameters were analysed to classify sediment waves and infer bottom-current characteristics across the study areas.

Three groups of sediment waves were identified: small-scale waves, predominantly found in the Argentine Basin; small to medium-scale waves occur in both the Argentine Basin and the Hatteras Outer Ridge; and large-scale sediment waves are characteristic of the Hatteras Outer Ridge. In the Argentine Basin, variation in bottom-current flow velocity reveals relatively high flow velocities in the southern Zapiola region and lower velocities in northern Zapiola region. Sediment-wave asymmetry was further employed to determine flow directions, indicating north-westward flow in northern Zapiola, south-westward flow in southern Zapiola, and predominantly southward flow over the Hatteras Outer Ridge.

These findings demonstrate that sediment-wave morphology serves as a reliable proxy for interpreting bottom-current strength and flow direction in deep-marine environments. The observed spatial variations in sediment-wave characteristics reflect differences in oceanographic conditions between the Argentine Basin and the Hatteras Outer Ridge, contributing to a better understanding of contourite depositional systems and deep-ocean circulation.

1. Introduction

Bottom-current induced sediment-waves are large-scale seabed bedforms developed through the interaction of persistent bottom current flows with contourite drifts in unconfined setting. These features range from a few meters to several kilometres in wavelength, meters in height (figure 1.), and are commonly found along continental margins, and contourite depositional systems. Their formation reflects the long-term influence of bottom-current processes on sediment transport, deposition, and erosion in the deep marine environment.

Because sediment-waves are closely linked to the strength, direction, and variability of bottom currents, their morphology serve as geological archives of past oceanographic conditions. One importance of sediment-waves studies is their ability to act as indicators of bottom-current activities and improves understanding of sediment transport processes within contourite systems.

The primary objective of this study is to investigate sediment-wave morphology and gain better understand bottom-current dynamics. Comparing sediment-waves from both areas allows for a better understanding of how different deep-water masses influence contourite systems.

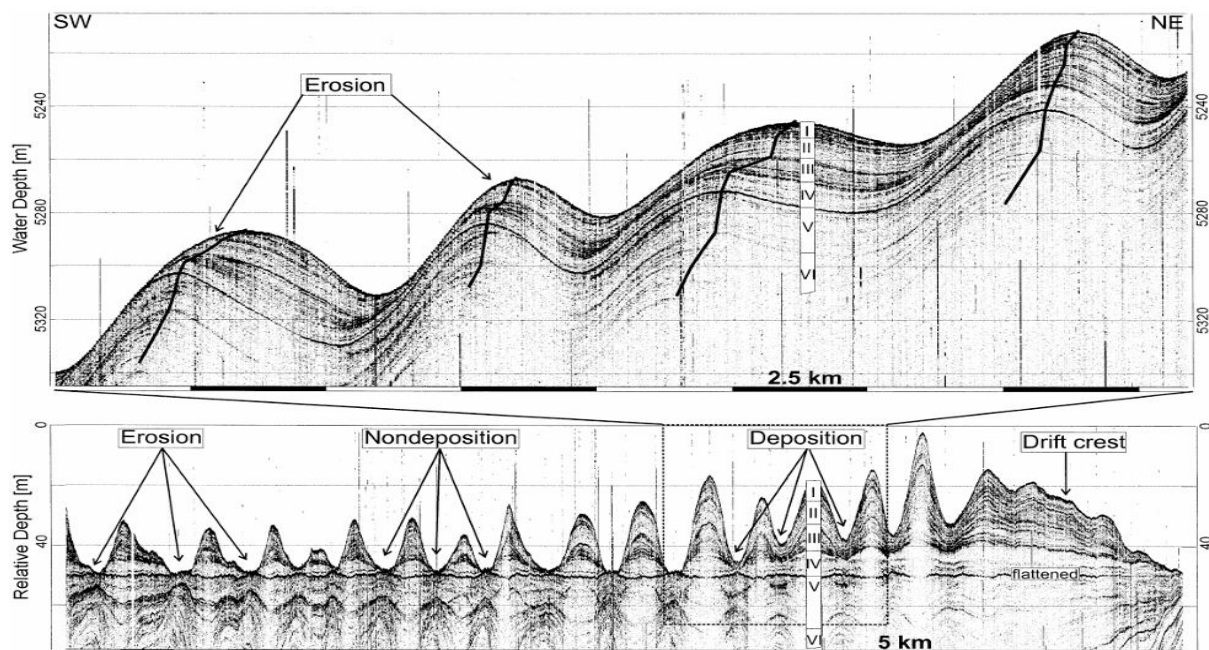


Figure 1. Cross-sectional view of sediment waves showing sediment waves deposited by bottom currents in the Argentine Basin (modified from von Lom-Keil, [Spieß](#) and Hopfauf, 2002).

2. Study Area

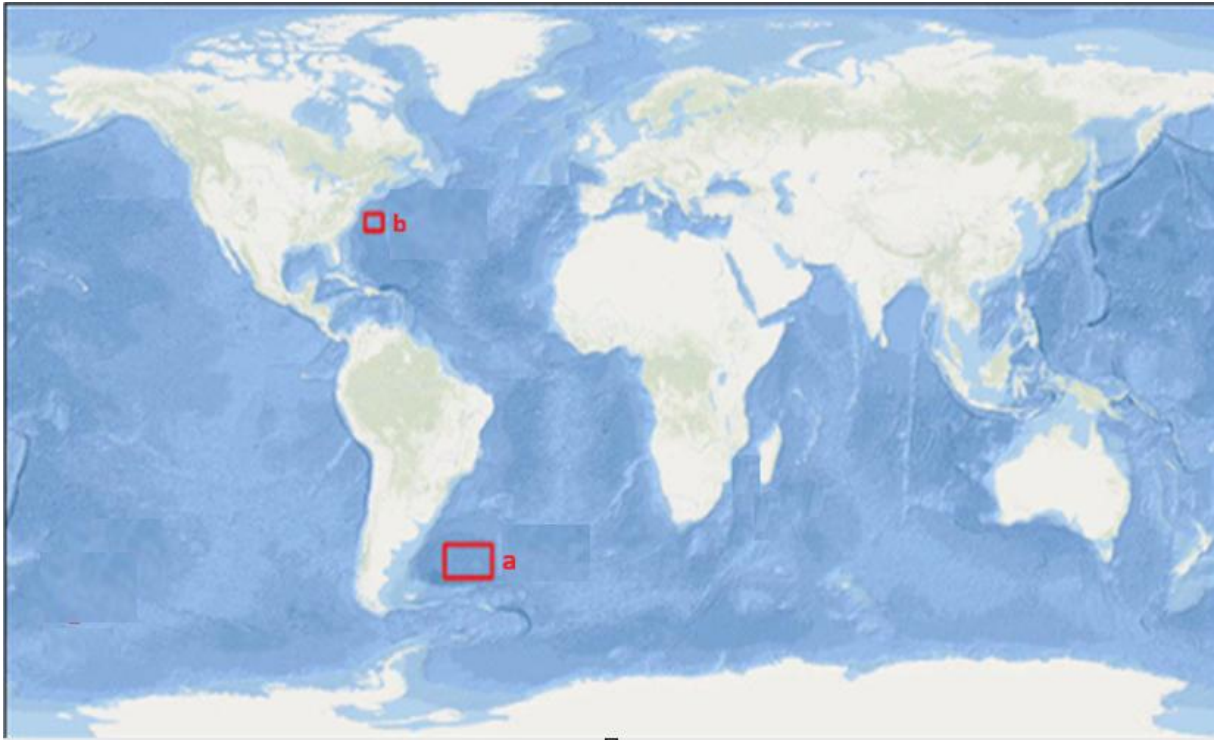


Figure 2. Study area showing locations of bottom current-induced sediment waves selected for the scope of this project. (c) Argentine Basin (b) Hatteras Outer Ridge.

This study focuses on two major contourite depositional systems in:

a. Zapiola Drift (Argentine Basin, 45°S, 45°W), located in the Southern Atlantic Ocean (figure 2.) represents one of the world's largest contourite depositional systems. The drift is characterized by extensive sediment waves developed under the influence of persistent bottom-current activity. Sediment transport and deposition are primarily controlled by the northward flowing Antarctic Bottom Water (AABW) along the Argentine margin. The resulting sediment-wave fields provide valuable evidence of long-term deep-water circulation and sedimentary processes in the South Atlantic Ocean.

b. Hatteras Outer Ridge (35°N, 74°W), situated offshore of Cape Hatteras along the eastern margin of North America in the western North Atlantic Ocean (figure 2.). This contourite drift system contains widespread bottom-current-generated sediment waves that have formed through the action of the southward flowing Western Boundary Undercurrent (WBUC). The interaction between the undercurrent and seafloor morphology promotes sediment redistribution and creates sediment-wave fields. These features preserve a record of variations in deep-water circulation and sediment dynamics in the North Atlantic Ocean.

3. Literature Research

3.1 Argentine Basin

Bottom currents play a fundamental role in sedimentation processes within the Argentine Basin. Evidence of their influence is commonly observed in seabed photographs and direct current measurement which have frequently revealed active bottom-current processes.

Previous studies have shown that bottom water enters the Argentine Basin primarily from the south from the East and West Falkland Gaps (Wüst, 1957; Le Pichon et al., 1971a; Georgi, 1981). These water masses flow westward along the base of the Falkland Escarpment before turning northward and following the Argentine continental margin (figure 3.) at depths greater than 3500 m. North of 35°S and at depths shallower than 4000 m, North Atlantic Deep Water flows southward along the margin (Reid et al., 1977).

The Argentine Basin is recognized as a region of high sea-surface variability and is susceptible to frequent benthic storms. These events generate strong, intermittent bottom currents with variable directions that can erode fine-grained sediments and resuspend them in the water column. Studies have demonstrated that such short energetic current events can significantly influence sediment reworking, redistribution, and deposition on the seafloor (Gardner and Sullivan, 1981; Hollister and McCave, 1984).

Multibeam echo sounding surveys have been used to study the distribution and morphology of mud waves, which are widely regarded as indicators of persistent contour-flowing bottom-currents. Mud waves generated by such currents are commonly oriented obliquely to regional bathymetric contours and generally migrate upslope (Flood, 1978; Lonsdale, 1983).

In the central Argentine Basin, sediment-wave studies indicate distinct migration patterns. In the north, preferential deposition occurs on the south flanks and erosion dominating the north flanks thereby causing upslope migration south-eastward. Similarly, the southern region exhibits upslope migration north-eastward, this is as a result of deposition on the northeastern flanks, and erosion concentrated on the southwestern flanks. These asymmetrical depositional and erosional patterns provide further evidence of sustained bottom-current activity and its role in shaping seabed morphology within the Argentine Basin.

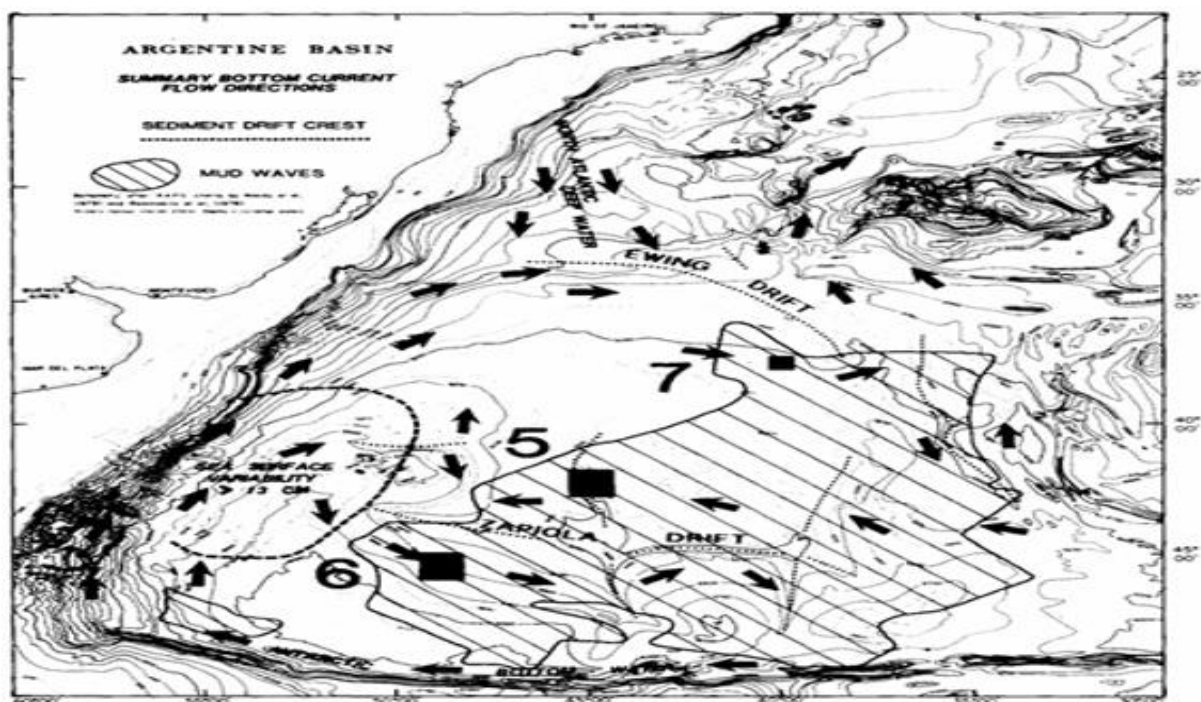


Figure 3. Bottom photograph showing current direction and strength in Argentine Basin.
FLOOD and SHOR. 1988

3.2. Hatteras Outer Ridge

The Hatteras Outer Ridge located on the lower continental rise to the east of Cape Hatteras, along the US continental margin, contains prominent field of large sediment waves (Figure 3.1). The development of these sediment waves has been largely controlled by persistent bottom-current activity, particularly the Western Boundary Undercurrent (WBUC), which flowed south-westward along the ridge (Brew, 1995; Gardner et al., 2015).

Sediment supply to the Hatteras Outer Ridge is primarily transport from the continental slope. A major contribution came from the Norfolk–Washington fan system, which acted as a local sediment source that was subsequently reworked by deep boundary currents during construction of the ridge (Locker and Laine, 1992). Once incorporated into the WBUC, these fine sediments were redistributed to form the extensive sediment-wave field.

Deposits transported through the Washington–Norfolk Canyon system partly infill the troughs separating individual sediment waves, while a smaller volume of sediment derived from the Hatteras Abyssal Plain accumulated within troughs along the southeastern margin of the ridge (Gardner et al., 2015).

Seismic evidence indicates that the sediment waves migrated upslope/ up current, and this pattern resulted from enhanced sediment accumulation on the northern, up current-facing flanks, leading to migration toward the ridge crest (Brew, 1995).

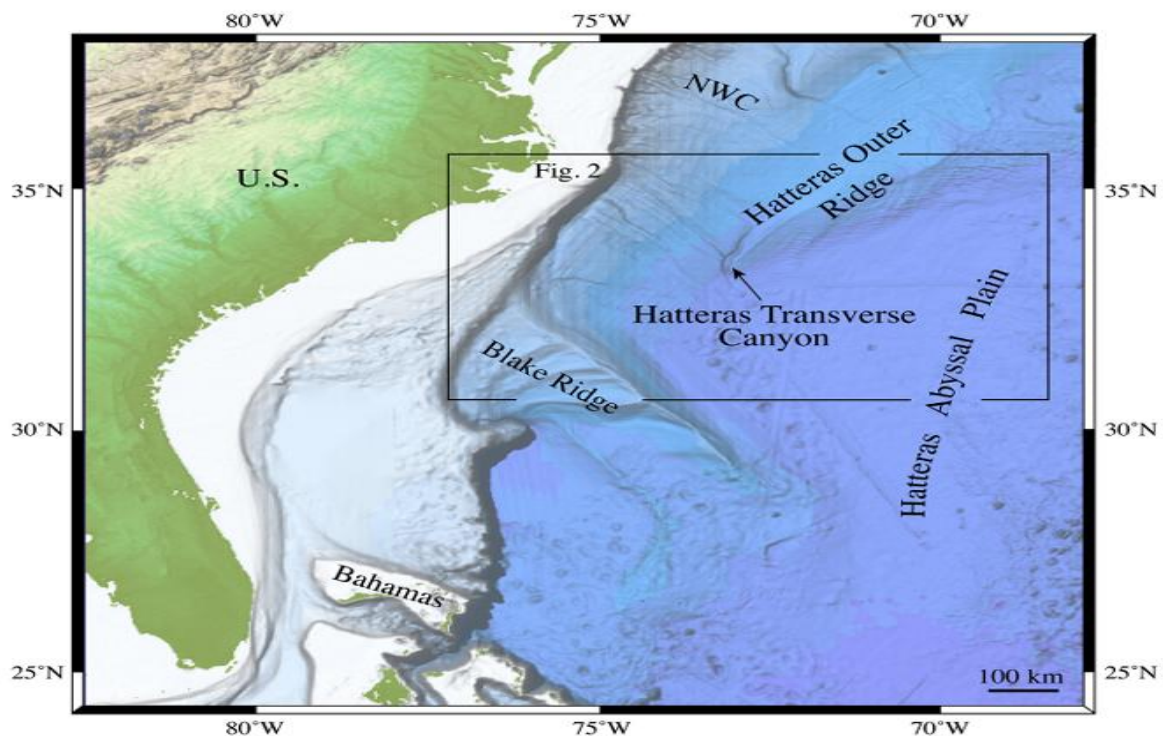


Figure 3.1 Location of Hatteras Transverse Canyon, Hatteras Outer Ridge and Hatteras Abyssal Plain on the U.S. Atlantic continental margin. Smith and Sandwell (1997, v. 17.1).

4. Research Questions

1. How does the morphology of sediment waves differ between Argentine Basin and Hatteras Ridge?
2. What is the relationship between sediment-wave morphology and current velocity in the study area?
3. How do sediment-wave morphologies reflect the hydrodynamic conditions responsible for their formation?

5. Methodology

The QGIS software tool was utilized for the project, and the following practical steps were taken.

6.1. Data Acquisition and Preparation: Bathymetric digital elevation model datasets of high resolution were obtained from the GEBCO and imported into QGIS for processing and analysis.

6.2. Coordinate Reference System Transformation: The original data in geographic coordinate reference system were reprojected into projected coordinate reference system to ensure accuracy for morphometric measurements.

6.3. Hillshade Generation: Hillshade were generated from the reprojected DEMs to enhance visualization morphological features.

6.4. Plugin Installation and Sediment-wave Mapping: After installing Terrain Profile Plugin Tool into QGIS, waves were identified and transects drawn across representative waves to extract morphological profiles. A total of five profiles were extracted from Argentine Basin and four from Hatteras Outer Ridge.

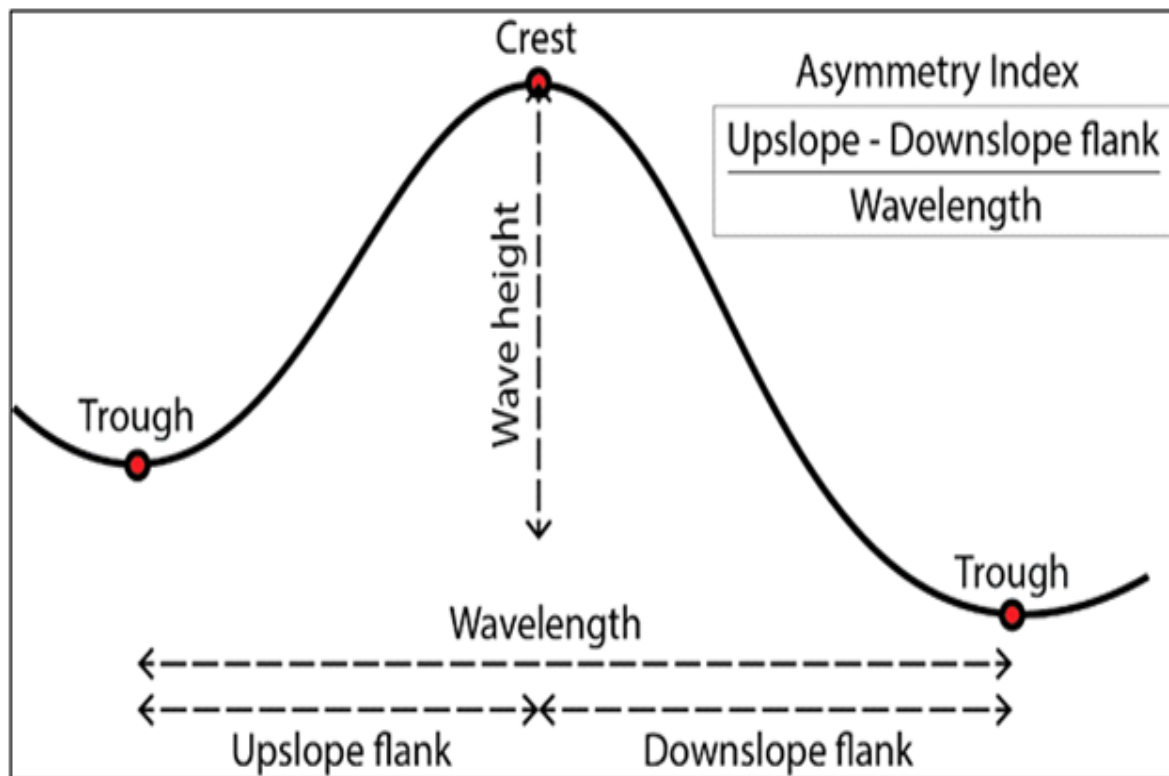


Figure 4. Schematic diagram showing measured morphometric parameters.

5.5. Morphometric Analysis

From the extracted sediment wave profiles, the following morphological parameters were extracted for this study and used to answer the research questions.

1. **Wavelength:** The horizontal distance between successive wave troughs, measured along profiles perpendicular to wave orientation (Figure. 4). This indicates flow characteristics such as persistence and energy of the current.
2. **Wave-height:** The vertical distance from the wave crest to the adjacent trough, representing the amplitude and intensity of sediment supply (Figure. 4).
3. **Depth:** The water depth at sediment wave field, recorded at the wave crest. This parameter is important for understanding the relationship between wave formation and oceanographic setting.
4. **Degree of Symmetry:** The comparative steepness of the stoss and lee sides of each wave. This parameter is obtained by measuring the horizontal distance between the wave crest and the trough on either side of the crest, subsequently, an asymmetry index is extracted by calculating the ratio.

Data Source:

GEBCO 2025 Grid (doi:10.5285/37c52e96-24ea-67ce-e063-7086abc05f29)

6. Chronological Project Narrative

Weeks \ Tasks	1	2	3	4	5	6	7	8	9	10	11	12
Literature research												
Data Acquisition												
Data processing												
Hillshade, Identification and wave extraction												
morphometric measurements												
Compilation of results												
Interpretation												
Report writing												

Figure 5. Chart of the timing/scheduling of the project.

The project was structured for 12-week period, and key tasks are contained in figure 5.

6.1. Challenges and Mitigation:

1. One of the primary challenges was managing the datasets for the analysis as the study involved working with multiple spatial datasets which often had different formats, and resolutions. Ensuring the right datasets were correctly imported, named, and prepared for analysis in QGIS required careful data management. Large raster datasets also increased processing time, but this was handled by reducing the data size.
2. Another challenge was dealing with coordinate reference system (CRS) inconsistencies. Datasets were downloaded in geographic coordinate reference systems, and this could not be used to for measurements as they were in degree. Resolving these issues required identifying the correct CRS for each dataset and reprojecting layers into a common coordinate system to ensure accurate mapping and analysis of sediment wave features.

7. Results

This section presents mapped distribution and measured characteristics of sediment waves across the study areas, highlighting spatial variability and key morphological differences.

7.1 Sediment wave mapped in Argentine Basin.

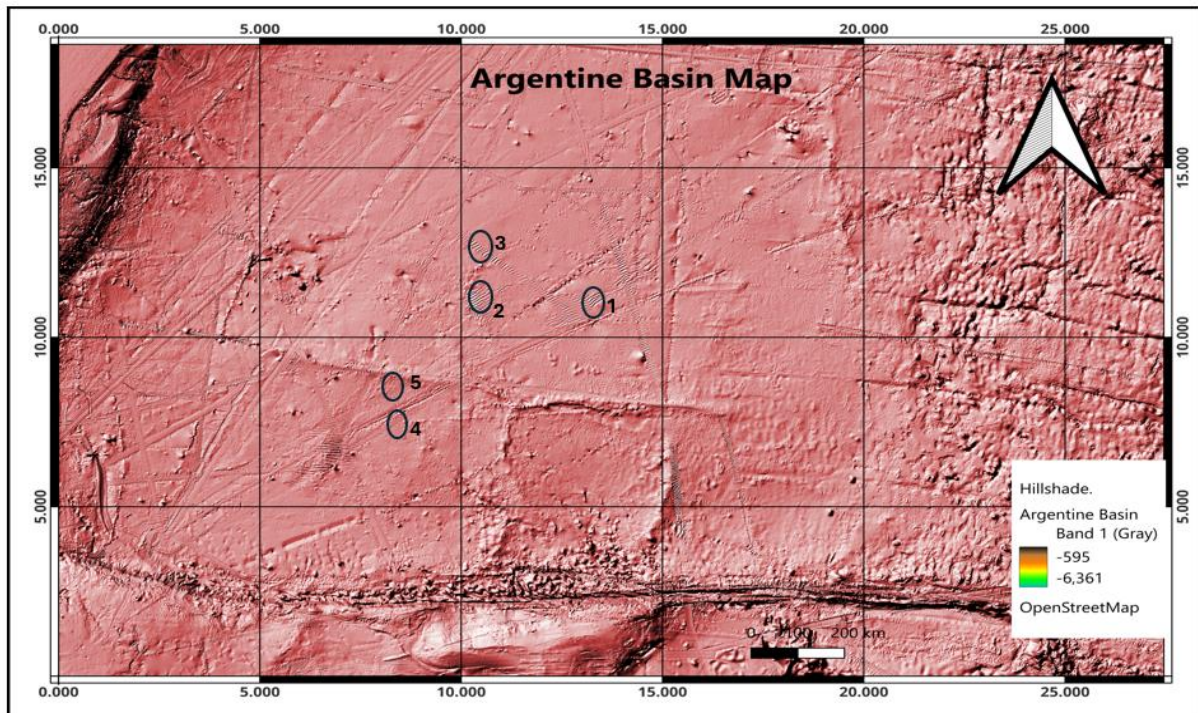


Figure 6. Map showing sediment wave study areas in Argentine Basin, the numbers represents locations.

Five sediment-wave profiles were identified and mapped across the Argentine Basin (Fig. 6). The northern sediment-wave fields (Profiles 1, 2, and 3) exhibit a northeast–southwest strike and are oriented in a northwest–southeast direction. The southern sediment-wave fields (Profiles 4 and 5) display a northwest–southeast strike and are oriented northeast–southwest. These distinct spatial patterns indicate regional variations in bottom-current regimes and sediment transport processes within the basin.

7.2 Sediment wave mapped in Hatteras Outer Ridge.

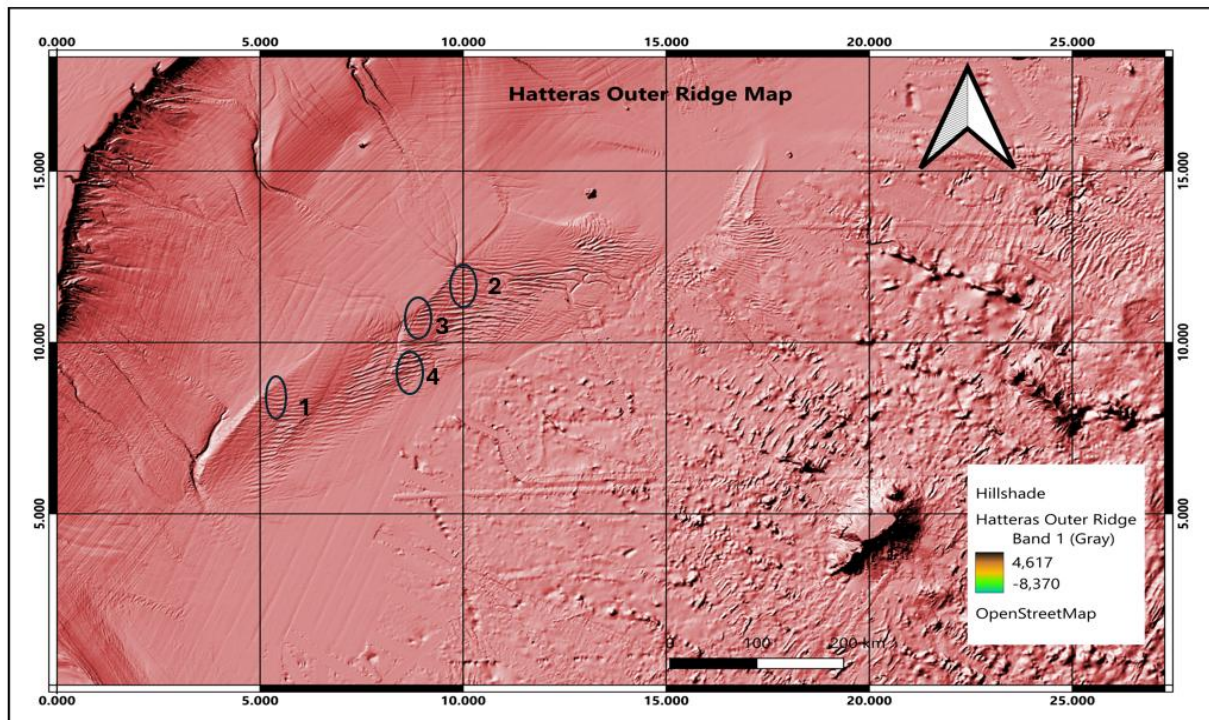


Figure 7. Map showing sediment wave study areas in Hatteras Outer Ridge, the numbers represents locations.

Four sediment-wave profiles were identified and mapped across the Hatteras Outer Ridge, as shown in Figure 7. All four profiles exhibit a consistent northeast–southwest strike and are oriented in a northwest–southeast direction. This uniform morphology suggests the influence of persistent bottom-current activity in shaping the sediment-wave field across the ridge.

7.3 Comparative Analysis of the Morphometric Characteristics.

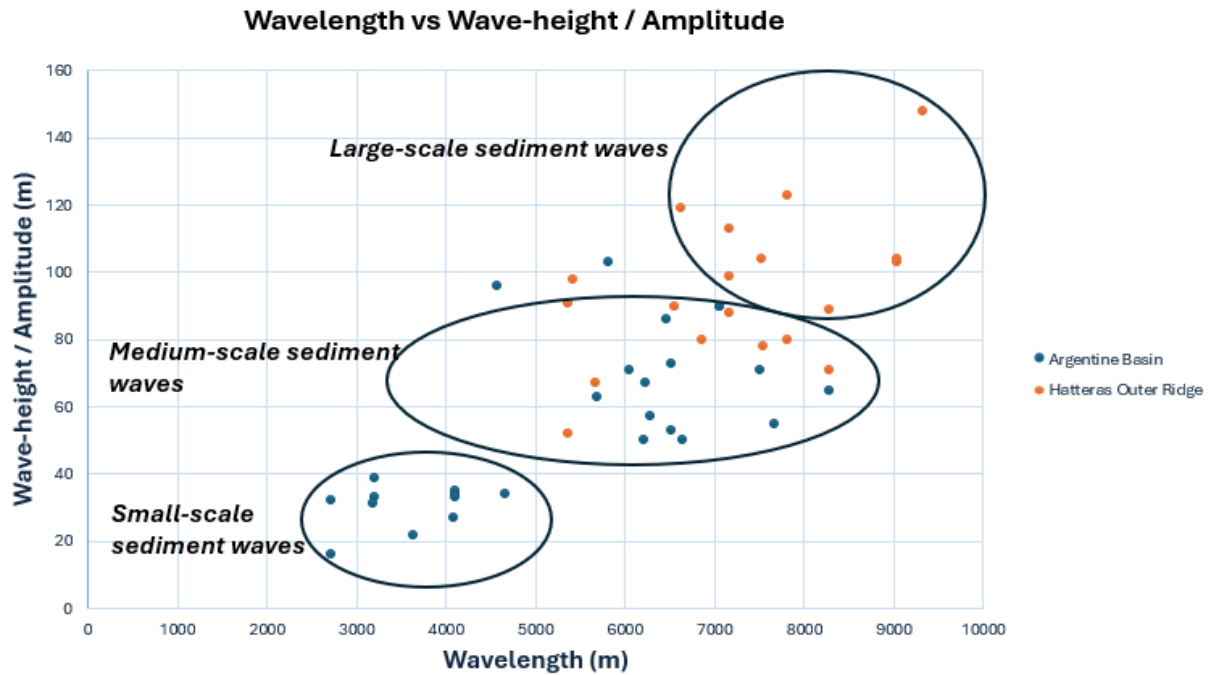


Figure 8. Scatter plot of wavelength and wave height from the study areas.

A comparative analysis of sediment-wave morphology in the Argentine Basin and the Hatteras Outer Ridge was conducted using wavelength and wave-height measurements as the primary classification parameters. The relationship between these morphological variables was examined using a scatter plot, which revealed the presence of three distinct sediment-wave classes across the two study areas.

Class 1 comprises the large-scale sediment waves observed exclusively in the Hatteras Outer Ridge. These features are characterized by wavelengths ranging from 6619.8 m to 9327.9 m and wave heights between 89 m and 148 m.

Class 2 represents medium-scale sediment waves occurring in both the Argentine Basin and the Hatteras Outer Ridge. This class exhibits wavelengths ranging from 5366.6 m to 8287.8 m and wave heights between 50 m and 91 m.

Class 3 consists of the small-scale sediment waves identified primarily within the Argentine Basin. These features are characterized by wavelengths ranging from 2723.4 m to 4657.6 m and wave heights between 16 m and 39 m.

The distribution of these classes indicates clear morphological differences between the two study areas. Sediment waves in the Hatters Outer Ridge generally exhibit larger dimensions, whereas the Argentine Basin is characterized by smaller and more variable bedforms. The overlap of medium-scale sediment waves in both regions suggests that similar depositional and hydrodynamic conditions may have influenced their development despite differences in their overall oceanographic settings.

7.4 Bottom-Current Velocity

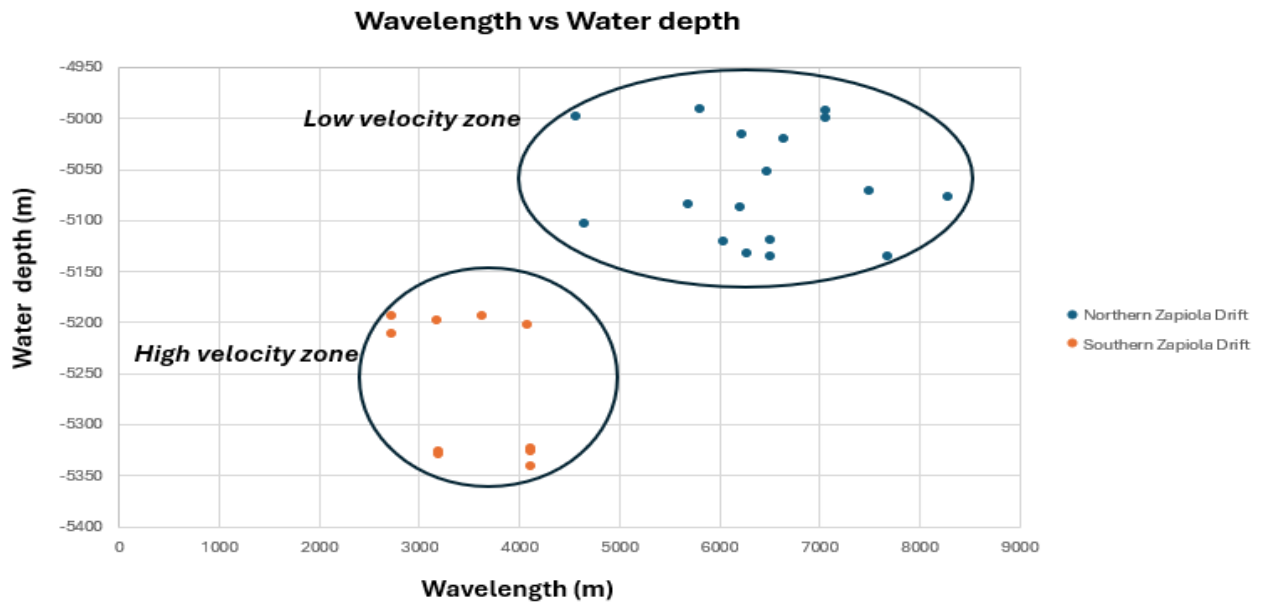


Figure 9. Scatter plot of wavelength and water depth from north and southern zapiola drift.

The distribution of bottom-current velocity zones within the Argentine Basin was inferred from the relationship between sediment-wave wavelength, used as a proxy for sedimentation rate, and water depth. A scatter plot of wavelength against water depth revealed a distinct velocity gradient, allowing the identification of zones characterized by relatively low and high bottom-current velocities.

The northern sector of the Zapiola region is characterized by relatively large sediment waves, with wavelengths ranging from 4565.1 m to 8220.1 m, occurring at comparatively shallower water depths of 4991 m to 5136 m. These larger wavelengths indicate higher rates of sediment accumulation and are interpreted as evidence of lower bottom-current velocities, which promote sediment deposition and the preservation of sediment-wave morphology.

In contrast, the southern Zapiola region exhibits smaller sediment waves, with wavelengths ranging from 2723.4 m to 4110.0 m, at greater water depths of 5193 m to 5341 m. The reduced wavelengths and deeper-water setting suggest lower sediment accumulation rates and are interpreted as indicators of stronger bottom-current velocities. These higher-energy currents likely enhance sediment erosion, transport, or inhibit deposition, resulting in the development of smaller sediment-wave features.

Overall, the observed relationship between wavelength and water depth suggests a transition from low-velocity depositional conditions in the northern Zapiola region to higher-velocity erosional or non-depositional conditions in the southern sector, reflecting spatial variations in bottom-current dynamics across the Argentine Basin.

7.5 Bottom-Current Flow Direction in The Study Areas.

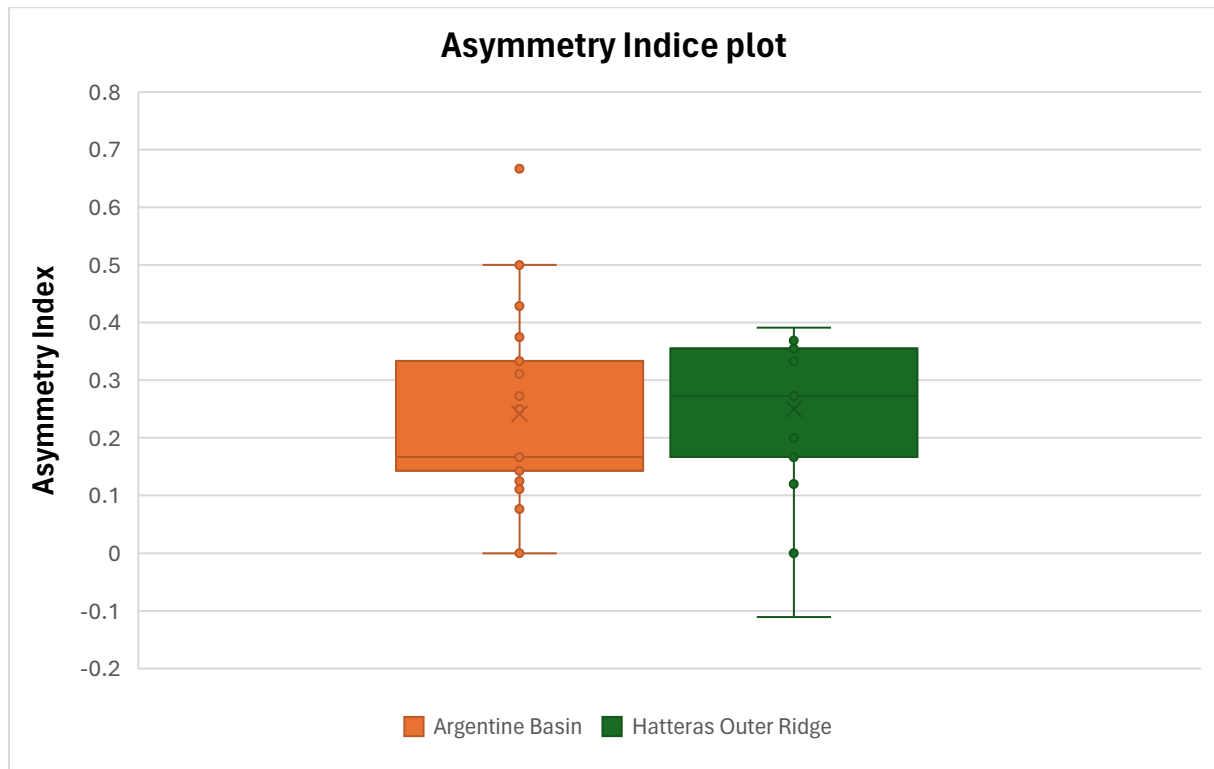


Figure 10. Box plot of the asymmetry Indices observed in the study areas.

Sediment-wave asymmetry was quantified using an asymmetry index calculated from the horizontal lengths of the wave flanks measured from left to right along each profile. The asymmetric index is defined as:

- $\text{Asymmetric Index} = (\text{Lleft} - \text{Lright}) / (\text{Lleft} + \text{Lright})$

Lleft = Horizontal length of the left flank.

Lright = Horizontal length of the right flank.

All analysed profiles exhibit predominantly positive asymmetry index values. In the Argentine Basin, Profile 1 shows values ranging from approximately -0.000000172 to 0.1538 , Profile 2 ranges from 0.2727 to 0.4999 , Profile 3 ranges from 0.1428 to 0.2857 , Profile 4 ranges from 0.111 to 0.3333 , and Profile 5 ranges from 0 to 0.6666 . These values collectively indicate a dominant asymmetry consistent with downslope flow directed towards the left, corresponding to a north-westward bottom-current flow direction in the northern Argentine Basin and a south-westward flow direction in the southern Argentine Basin.

In the Hatteras Outer Ridge, Profile 1 exhibits asymmetry values ranging from 0.2727 to 0.3913, Profile 2 ranges from 0 to 0.3636, Profile 3 ranges from 0.1666 to 0.3687, and Profile 4 ranges from -0.111 to 0.3548. The generally positive asymmetry values in this region similarly indicate a dominant downslope flow direction towards the left, corresponding to a southward bottom-current flow direction across the study area.

The distribution of asymmetry indices across both regions suggests consistent but spatially varying bottom-current influence, with sediment-wave morphology reflecting regional differences in flow direction and intensity.

7.5.1. Wave Analysis and Bottom-Current Direction for Profile 1 in the Argentine Basin.

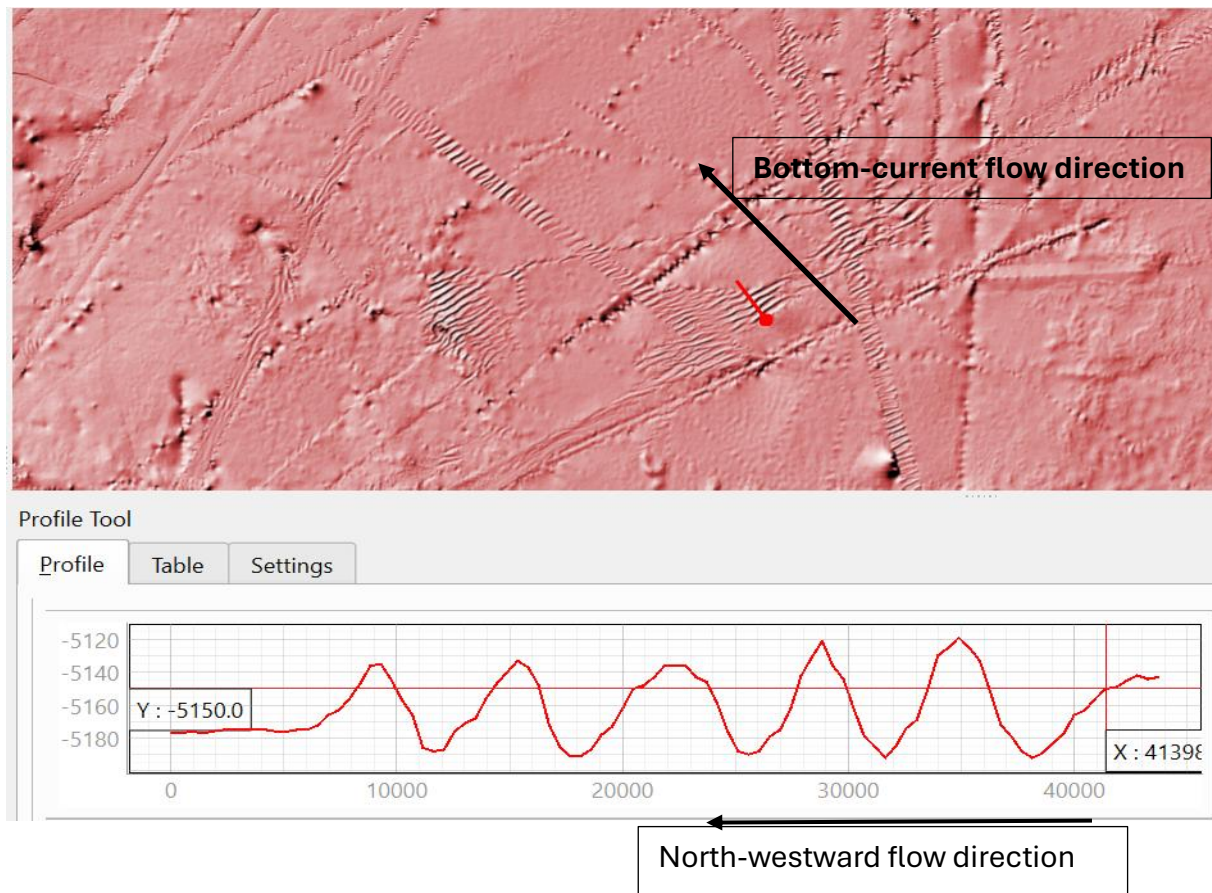


Figure 11. Morphometric Profile showing flow direction of sediment waves for profile 1 in Argentine Basin.

The first sediment wave mapped in northern zapiola drift is has 5 wave forms that are slightly variable. They strike northeast–southwest, orientated northwest-southeast, and the asymmetric index shows north-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 6046.945m (4) – 7674.969m (3)
2. Wave-height : 53m (1) – 73m (5)
3. Water depth: -5119m (5) - (-5136m) (3)
4. Asymmetry index: -0.000000172 (4) – 0.1538 (2)

Each complete wave in this profile is numbered from the left where measurement started (first wave :1, second wave :2, third wave: 3, fourth wave: 4, fifth wave: 5).

Note: The same numbering procedure is applied for other profiles in subsequent locations in this study.

7.5.2. Wave Analysis and Bottom-Current Direction for Profile 2 in the Argentine Basin.

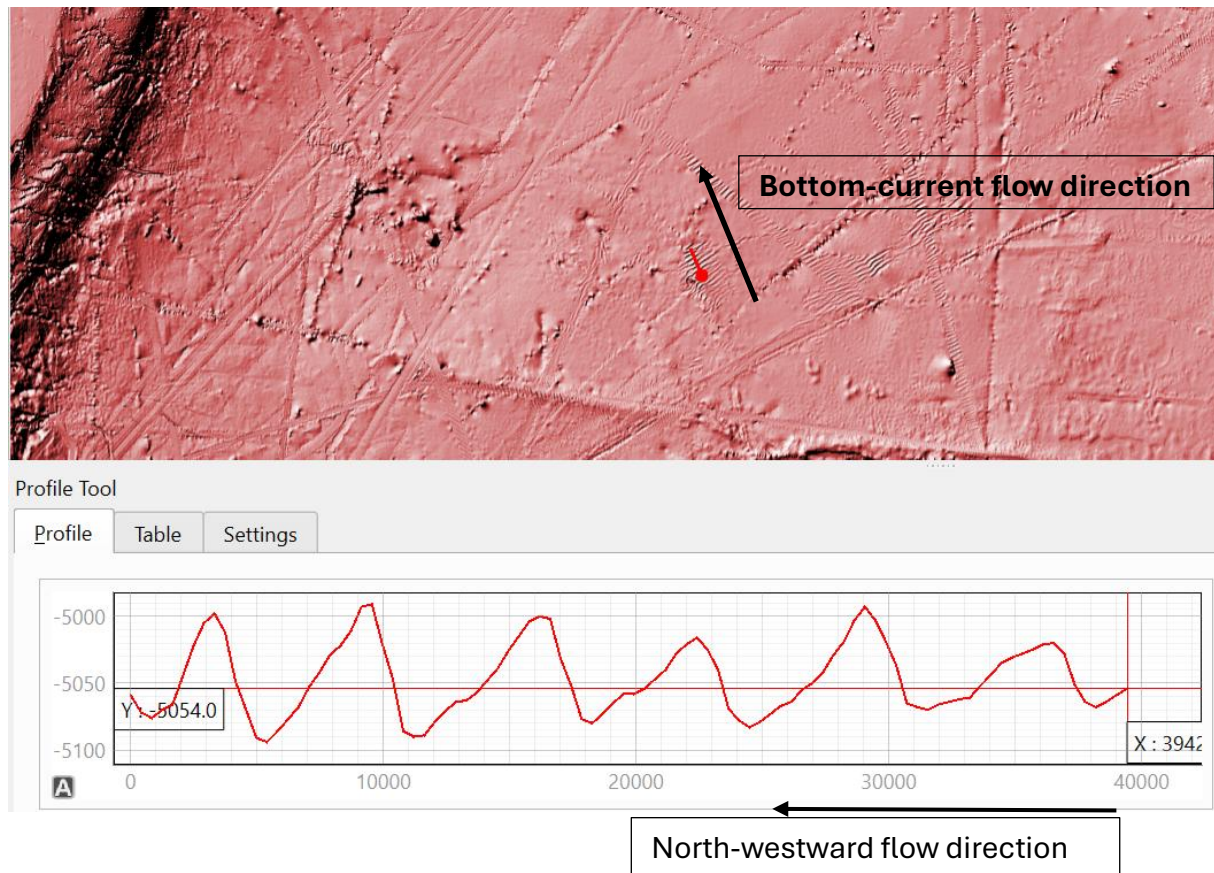


Figure 12. Morphometric Profile showing flow direction of sediment waves for profile 2 in Argentine Basin.

Profile 2 of Zapiola drift located in the north, further west of profile 1 in northern Zapiola has 6 wave forms with variable sizes and shapes. They strike northeast–southwest, orientated northwest-southeast, and the asymmetric index shows north-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 4565.118m (1) – 7055.183m (3, 5)
2. Wave-height: 50m (6) – 103m (2)
3. Water depth: -4991m (2) – (-5020m) (6)
4. Asymmetry index: 0.2727 (1) – 0.4999 (3)

7.5.3. Wave Analysis and Bottom-Current Direction for Profile 3 in the Argentine Basin.

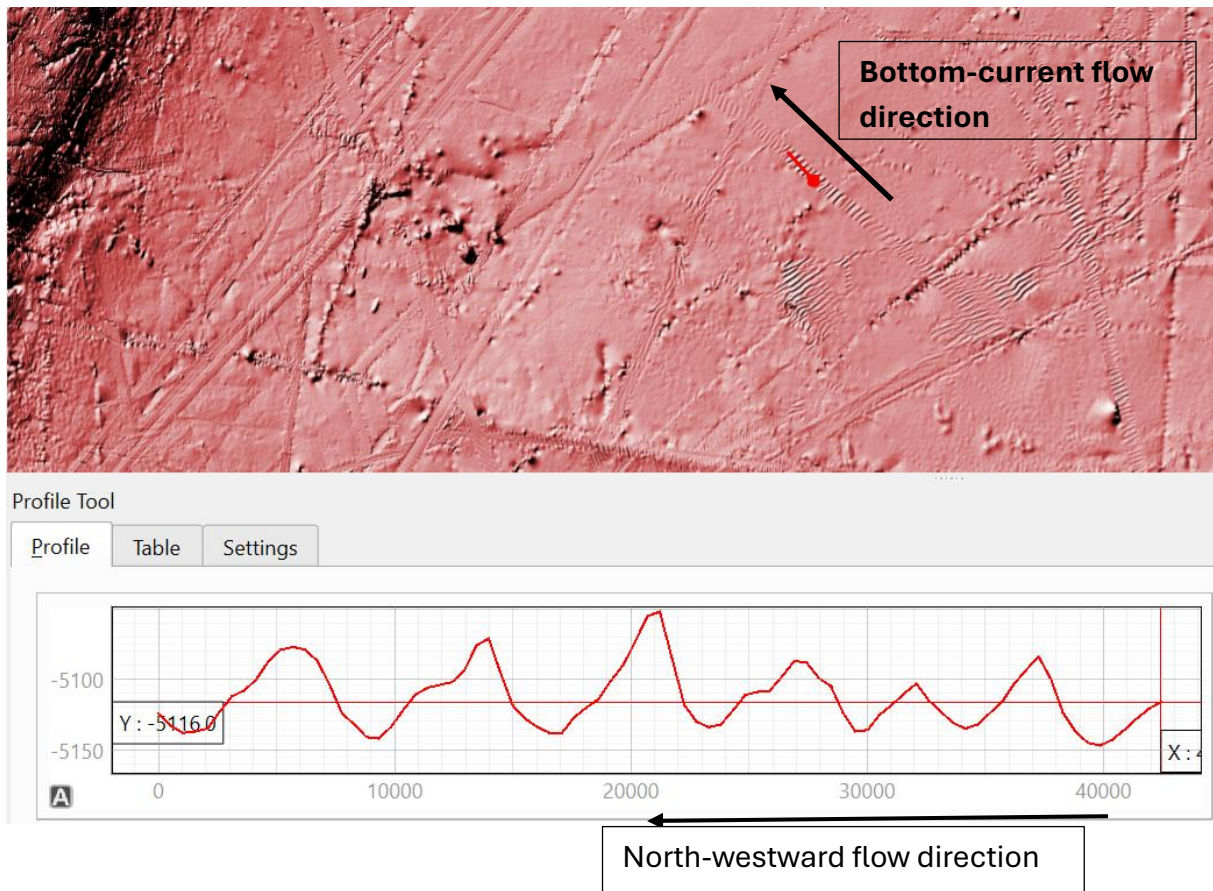


Figure 13. Morphometric Profile showing flow direction of sediment waves for profile 3 in Argentine Basin.

Profile 3 in north zapiola has 5 wave forms that strike northeast–southwest, orientated northwest-southeast, and the asymmetric index shows north-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 4657.554m (5) – 8280.095m (1)
2. Wave-height: 34m (5) – 86m (3)
3. Water depth: -5052m (3) – (-5103m) (5)
4. Asymmetry index: 0.1428 (1) – 0.2857 (2)

7.5.4. Wave Analysis and Bottom-Current Direction for Profile 4 in the Argentine Basin.

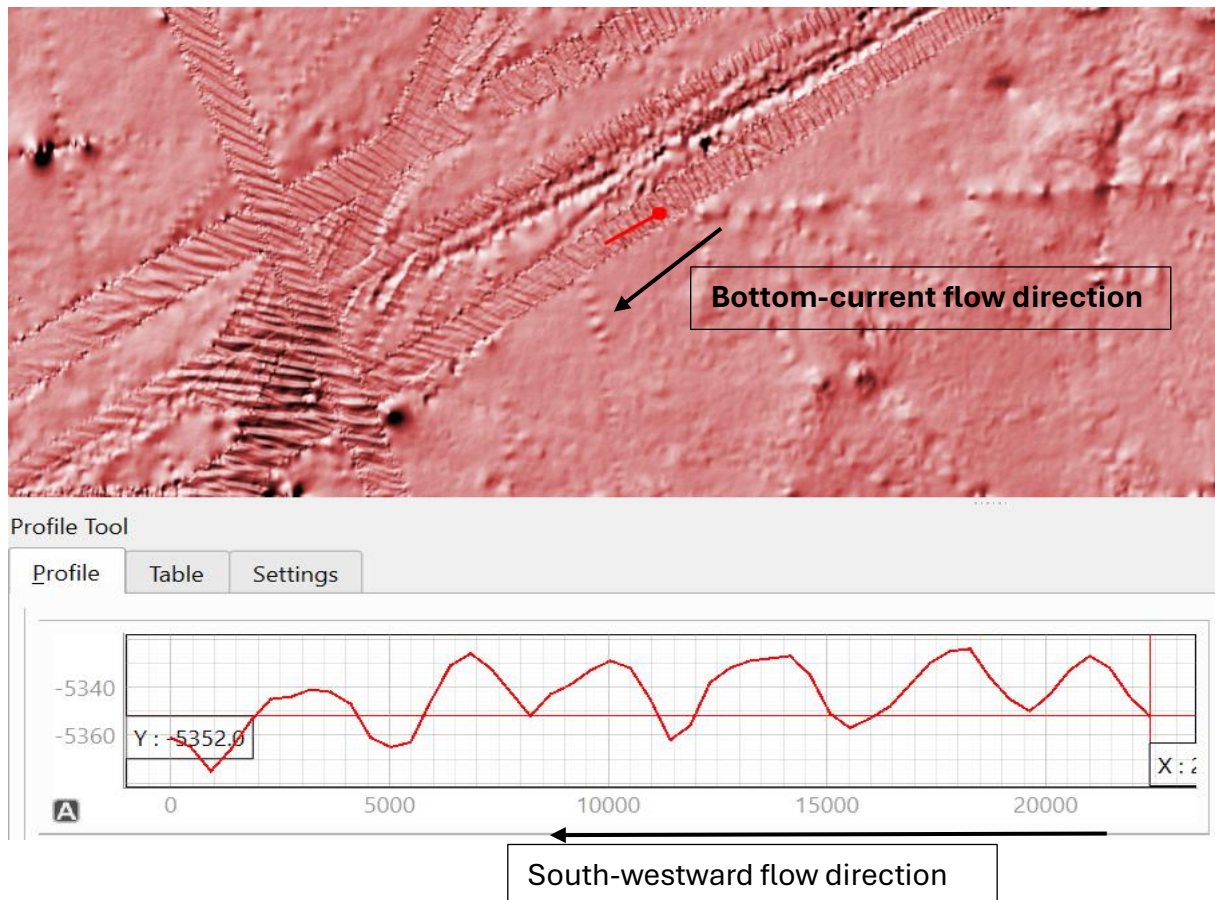


Figure 14. Morphometric Profile showing flow direction of sediment waves for profile 4 in Argentine Basin.

Profile 4 in southern zapiola strike northwest-southeast, oriented northeast-southwest, and the asymmetric index values shows south-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 3196.703m (2,3) – 4110.047m (4, 5)
2. Wave-height: 33m (3, 5) – 39m (2)
3. Water depth: -5326m (2) – (-5341m) (1)
4. Asymmetry index: 0.111 (1) – 0.3333 (4, 5)

7.5.5. Wave Analysis and Bottom-Current Direction for Profile 5 in the Argentine Basin.

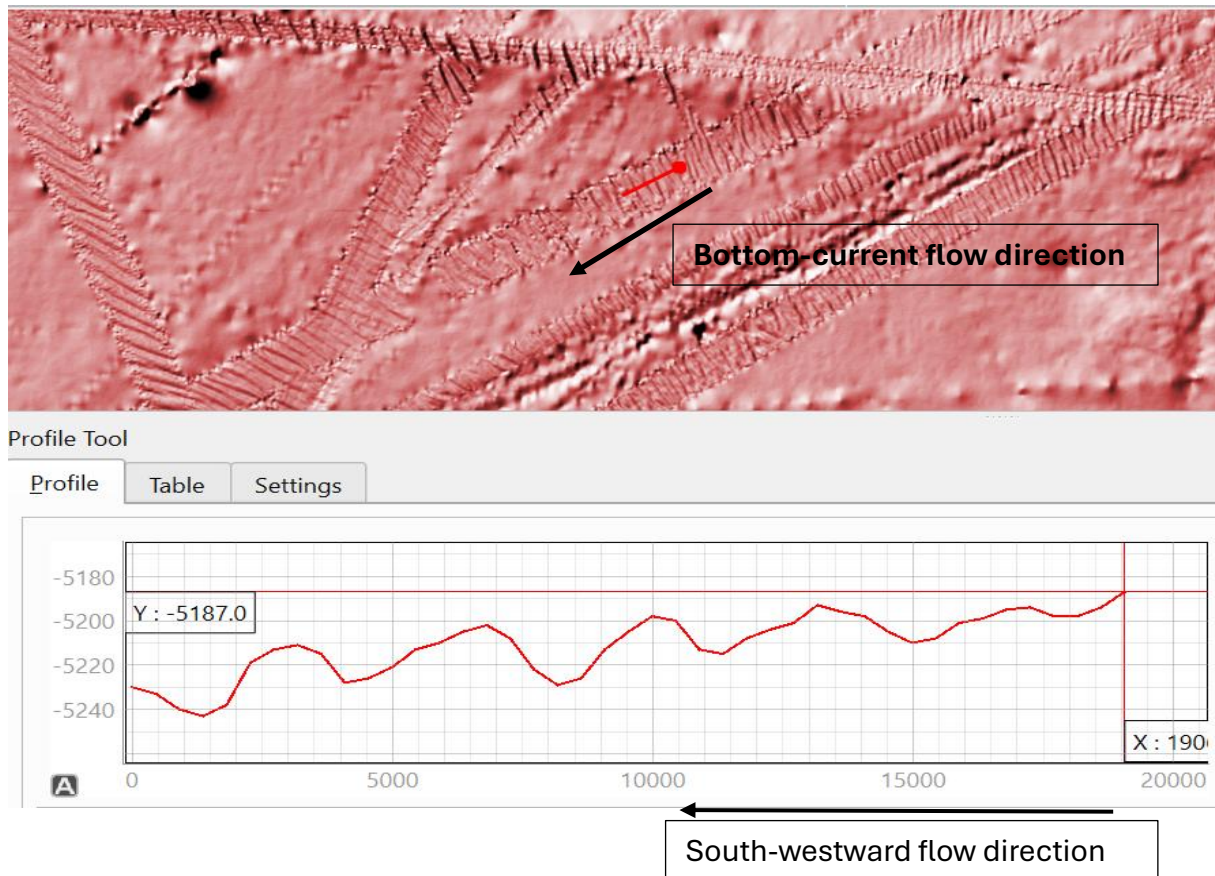


Figure 15. Morphometric Profile showing flow direction of sediment waves for profile 5 in Argentine Basin.

Profile 5 in southern zapiola strike northwest-southeast, oriented northeast-southwest, and the asymmetric index values shows south-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 2723.355m (1) – 4085.032m (2)
2. Wave-height: 16m (5) -32m (1)
3. Water depth: -5193m (4) – (-5211m) (1)
4. Asymmetry index: 0 (4) – 0.6666 (5)

7.5.6. Wave Analysis and Bottom-Current Direction for Profile 1 in Hatteras Outer Ridge.

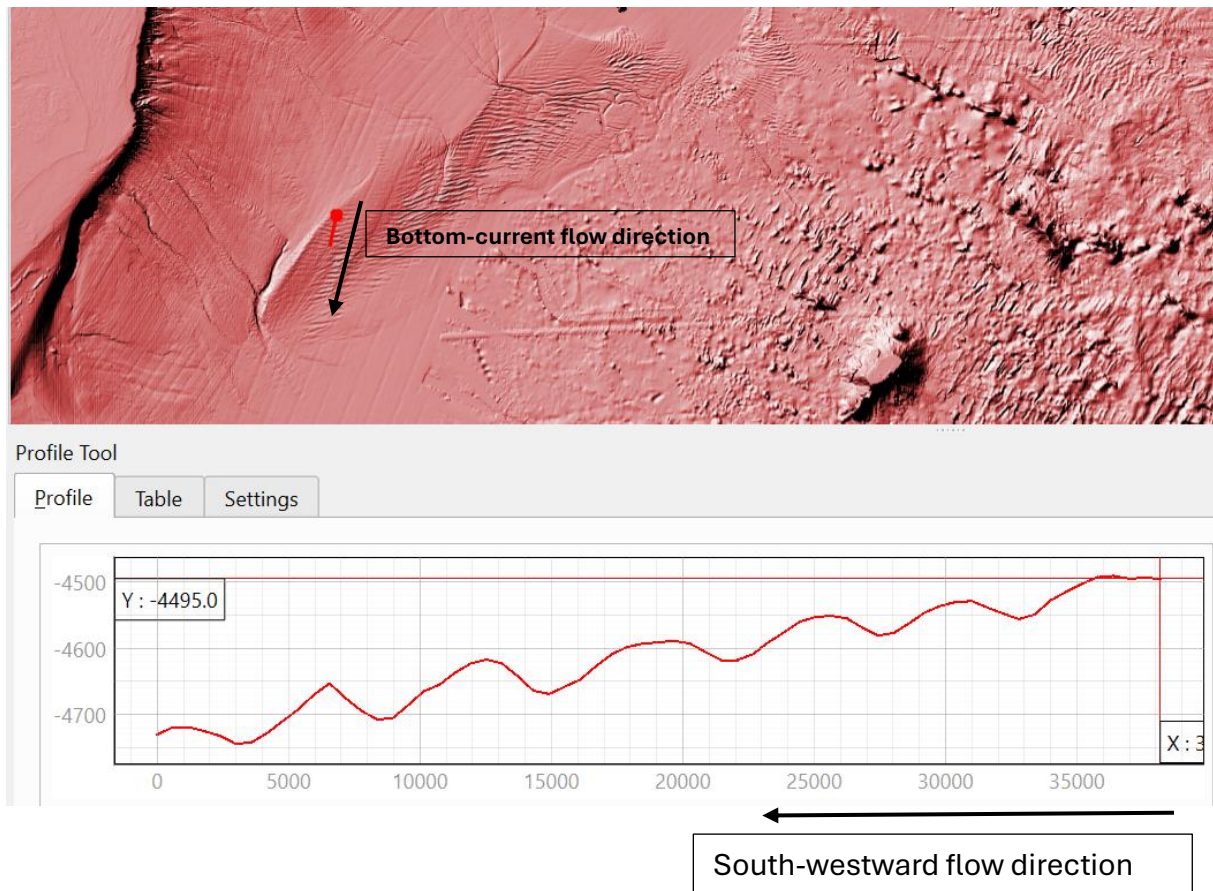


Figure 16. Morphometric Profile showing flow direction of sediment waves for profile 1 in Hatteras Outer Ridge.

The first profile in Hatteras outer ridge is located south of the ridge, strike northeast-southwest, oriented northwest-southeast, and the asymmetric index value show south-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 5366.647m (1,5) – 6857.383m (3)
2. Wave-height: 52m (5) – 91m (1)
3. Water depth: -4529m (5) – (-4653m) (1)
4. Asymmetry Index: 0.2727 (2) – 0.3913(3)

7.5.7. Wave Analysis and Bottom-Current Direction for Profile 2 in Hatteras Outer Ridge.

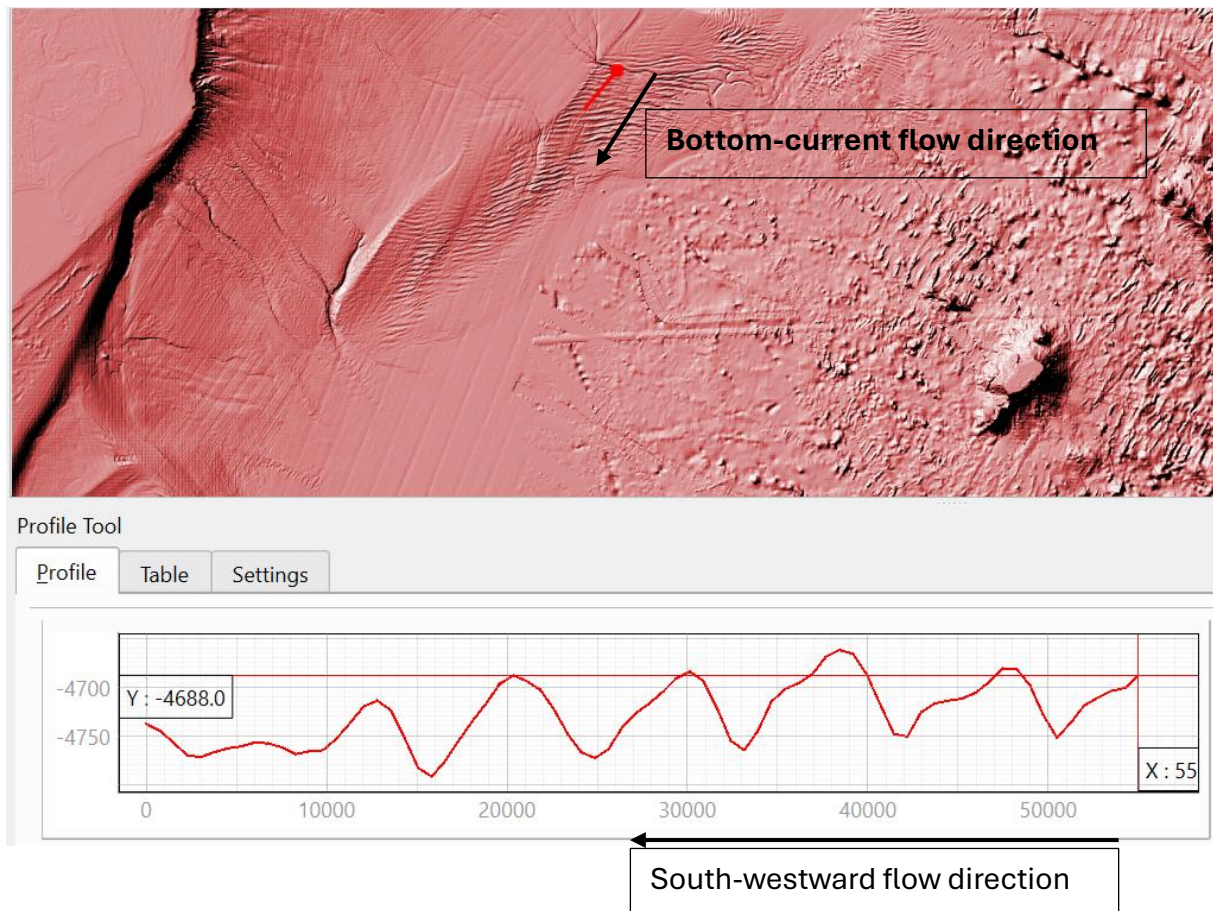


Figure 17. Morphometric Profile showing flow direction of sediment waves for profile 2 in Hatteras Outer Ridge.

The second profile of Hatteras Outer Ridge is located northernmost of the area, strike northeast-southwest, oriented northwest-southeast, and the asymmetric index value show south-westward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 7534.403m (1) - 9041.284m (2,4)
2. Wave-height: 71m (5) - 104m (2)
3. Water depth: -4662m (4) - (-4714m) (1)
4. Asymmetry Index: 0 (2) - 0.3636 (5)

7.5.8. Wave Analysis and Bottom-Current Direction for Profile 3 in Hatteras Outer Ridge.

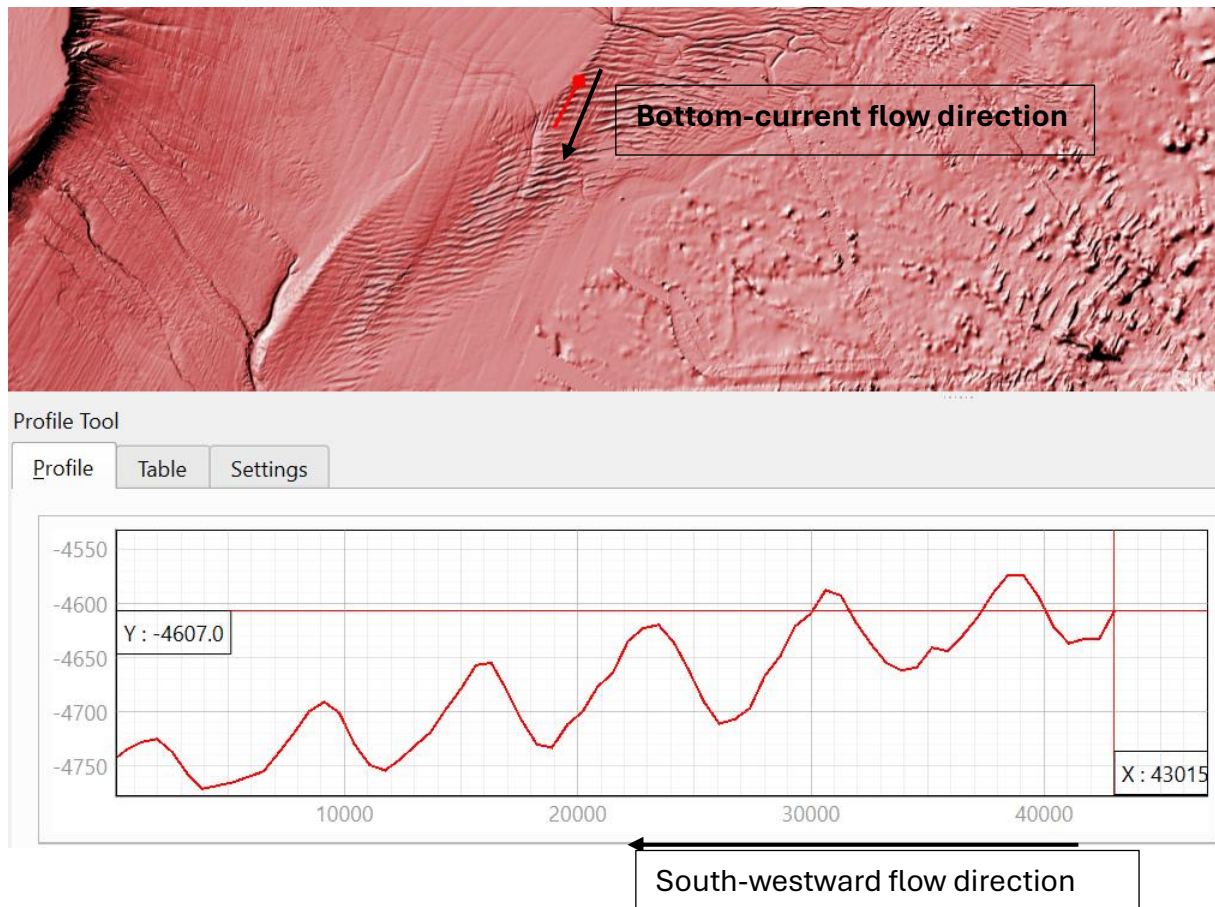


Figure 18. Morphometric Profile showing flow direction of sediment waves for profile 3 in Hatteras Outer Ridge.

Located 3 on Hatteras outer ridge strike northeast-southwest, oriented northwest-southeast, and the asymmetric index value show south-westward bottom-current flow direction.

The morphological analysis:

1. Wavelength: 7169.33m (2,3,5) – 7821.087m (1,4)
2. Wave-height: 80m (1) – 123m (4)
3. Water depth: -4574m (5) – (-4691m) (1)
4. Asymmetry: 0.1666(4) - 0.3687 (5)

7.5.9. Wave Analysis and Bottom-Current Direction for Profile 4 in Hatteras Outer Ridge.

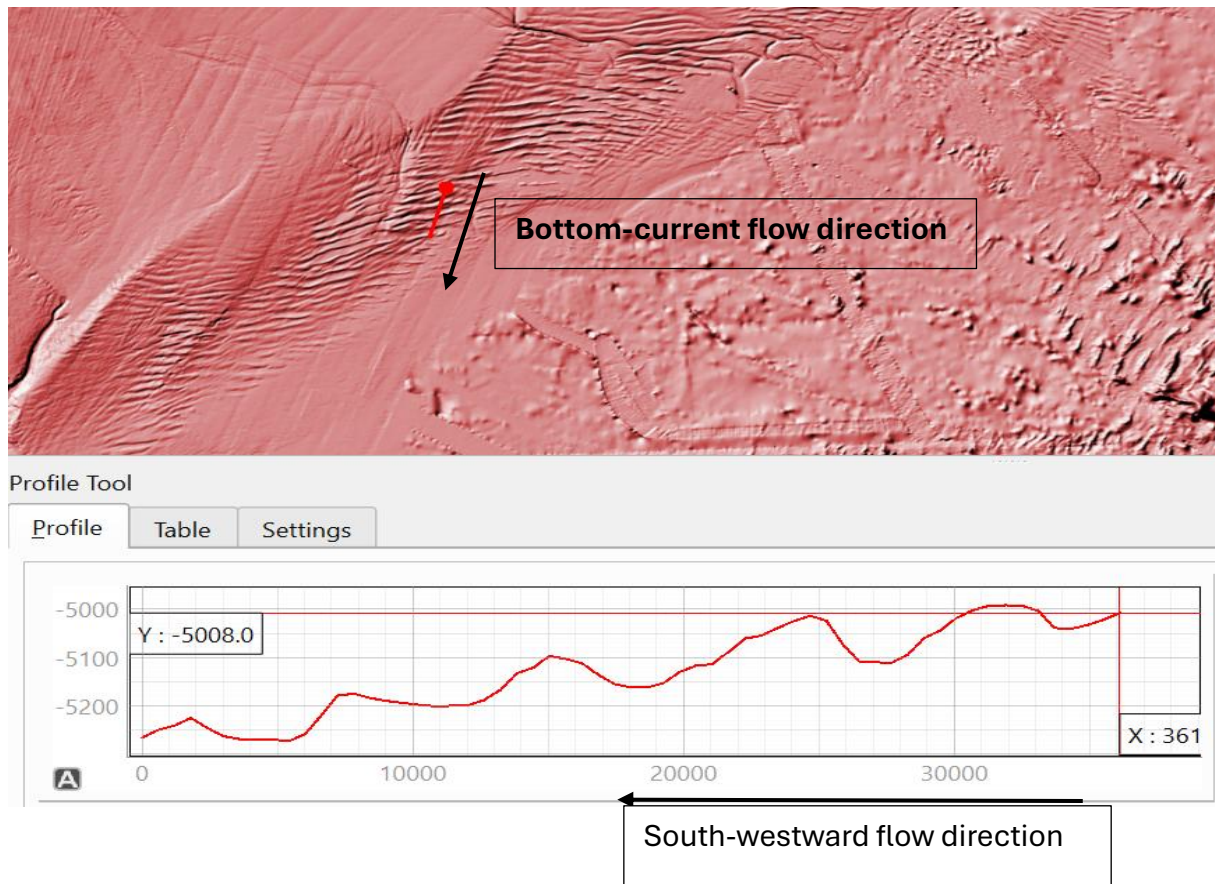


Figure 19. Morphometric Profile showing flow direction of sediment waves for profile 4 in Hatteras Outer Ridge.

Profile 4 in Hatteras outer ridge strike northeast-southwest, oriented northwest-southeast, and the asymmetric index value show southward bottom-current flow direction.

The morphological parameters measured include:

1. Wavelength: 5416.195m (1) – 9327.892m (3)
2. Wave-height: 98m (1) – 148m (3)
3. Water depth: -4992m (4) – (-5175m) (1)
4. Asymmetry: -0.111 (1) – 0.3548 (3)

8. Discussion

This study utilized QGIS sediment wave mapping and morphological analysis to investigate sediment wave characteristics within the Argentine Basin and the Hatteras Outer Ridge. The analysis focused on classifying their size and distribution, determining flow directions, and estimating variations in bottom-current velocity.

The results demonstrate clear differences in sediment wave characteristics between the two study areas. Sediment waves identified in the Argentine Basin were mainly classified as small-scale to small-medium scale and occurred at greater water depths. In contrast, sediment waves on the Hatteras Outer Ridge were classified as medium to medium-large scale features and developed in a relatively shallower environment. These differences suggest that sediment wave morphology is strongly influenced by environmental conditions, including water depth, sediment supply, and variations in bottom-current energy. The larger sediment waves observed on the Hatteras Outer Ridge may reflect more favourable conditions for sediment accumulation compared with the deeper Argentine Basin setting.

The asymmetry index provided important information about sediment transport processes. Predominantly positive asymmetry indices were observed in both regions, indicating a consistent downslope bottom-current flow activity rather than random sediment redistribution. However, the inferred flow orientations differed between the locations. In the Argentine Basin, the North Zapiola Current was associated with a north-westward flow direction, while the South Zapiola Current showed a south-westward orientation. Similarly, sediment waves on the Hatteras Outer Ridge indicated a south-westward flow direction. These variations reflect the influence of regional oceanographic circulation patterns on sediment transport pathways and demonstrate that sediment wave morphology can be used as an indicator of past and present bottom-current direction.

The bottom-current velocity analysis of the Argentine Basin revealed spatial variations in current strength. The northern part of Zapiola drift was inferred as a lower-velocity region, whereas the southern part showed characteristics associated with higher bottom-current velocities. This variation may explain differences in sediment wave development, as stronger velocities can enhance erosion or non-deposition, while low velocities promote deposition. However, the velocity estimates are based on morphological interpretations using the model by Hanno von Lom-Keil et.al 2002 and should be considered as relative indicators rather than direct measurements.

Overall, the results highlight the importance of geomorphological indicators to reconstruct bottom-current patterns.

9. Conclusion

This study highlights the usefulness of sediment wave morphology as a proxy for investigating deep-marine current systems. The combination of GIS mapping, morphometric analysis, and interpretation of sediment wave characteristics provides valuable insights into the relationship between ocean currents, sediment transport, and seabed morphology. Future work incorporating direct oceanographic measurements, higher-resolution bathymetric data, and sediment property analysis would improve the accuracy of bottom-current reconstructions and provide a more complete understanding of sediment wave evolution.

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